How to read papers?

by Briane Paul V. Samson, PhD

Adapted from the Introduction to Research slides of KIXLab

Anatomy of a research paper

- Title
- Authors
- Abstract
- Introduction
- Background or Related Work
- Main Technique or Theory
- Research Questions
- Experimental Setup
- Results
- Discussions/Future Work
- Threats to Validity (Limitations)
- Conclusion
- Acknowledgement
- References
- Appendix

Title

What the paper is about

- Contribution
- Domain area
- Technique
- Study Area
- Catchy title

Authors

Signal for paper quality

- What is their expertise?
- How long have they been doing this?

Abstract

- Short summary of the entire paper
- Around 250-300 words
- Enough to get the gist of the key contributions

Introduction

- A slightly longer summary of the entire paper
- Usually contains the following:
  - Status quo
    - What is the problem and why should we solve it?
  - Past work and state-of-the-art
    - What have others tried before?
    - What are their limitations? What are still open questions?
  - Paper's approach
    - How are we doing things differently?
  - Intended contributions
  - Organization of the rest of the paper

Background/Related Work

- A synthesis of existing works related to your RQ
- Background – Describing existing materials
- Related Works – Comparison of existing works to motivate yours
- Positioning your work
  - Are you tackling the same problem or a different one?
  - Are you trying a different approach?
  - Are you trying to improve or optimize a certain outcome?

Threats to Validity/Limitations

- Honest discussion of factors that may affect the validity of your conclusions
- Internal validity: Have you evaluated your solution correctly?
  - Control of certain variables to make sure effects are caused by the solution
- External validity: How well does your claim generalize?
  - How representative are your participants? Are there no bias?
- Construct validity: Are you measuring what you are supposed to measure?

Research Areas

- artificial intelligence
- computer systems and networks
- security
- database systems
- human-computer interaction
- vision and graphics
- numerical analysis
- programming languages
- software engineering
- bioinformatics
- theory of computing

What is computer science? https://undergrad.cs.umd.edu/what-computer-science

How to read papers *efficiently*?

Three-pass Approach

1st pass

- Quick scan
- Decide whether to read for more passes

2nd pass

- Read with greater care
- Not yet the full details

3rd pass

- Fully understand
- Re-do the steps on your mind

Keshav, S. (2007). How to read a paper. *ACM SIGCOMM Computer Communication Review*, *37*(3), 83-84.

First pass

1. Read the title, abstract and introduction
2. Read the section and sub-headings, but ignore everything else
3. Read the conclusion
4. Scan the references

1. Category

   What type of paper is this?

2. Context

   Which other papers is it related to? Which theoretical bases were used to analyze the problem?

3. Correctness

   Do the assumptions appear to be valid?

4. Contributions

   What are the main contributions?

5. Clarity

   Is the paper well-written?

What is computer science? https://undergrad.cs.umd.edu/what-computer-science

Questions?